

# AFM Journal Search Tool v2

Integrated tool for AFM Manual Introduction writing:

Journal search · Quality scoring · PDF collection · ZIP export

v2.0 Final

Park Systems

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English Edition

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## Tool Overview

### 1.1 What is AFM Journal Search Tool?

AFM Journal Search Tool v2 is an integrated research tool designed to systematize the writing of **Chapter 1 Introduction** in AFM (Atomic Force Microscopy) measurement mode manuals.

The tool consolidates journal searching, evaluation, and organization — previously performed differently by each researcher — into a **single HTML file**. It runs instantly in a browser with no installation required and can be shared across the entire team as one file.

#### Problems This Tool Solves

Inconsistent search criteria across researchers, no unified evaluation of IF · citations · equation count, and the repeated cycle of keyword collection → search → notes → draft writing — all solved in one place.

### 1.2 Key Features

#### Single HTML File

No installation required. Opens instantly in Chrome/Edge. Most features work offline.

#### API Auto-Search

Uses CrossRef API + Unpaywall API to automatically search journals by keyword and verify PDF accessibility in real time.

#### ★ Auto Quality Scoring

Automatically calculates a 0–100 quality score based on Impact Factor, citations, equation count, and physics/circuit content.

#### Batch ZIP Export

Download metadata, report, draft template, checklist, and open-access PDFs all in one ZIP file.

### 1.3 System Requirements

| Item          | Minimum                    | Recommended                             |
|---------------|----------------------------|-----------------------------------------|
| Browser       | Chrome 90+ / Edge 90+      | Latest Chrome                           |
| Internet      | Required for auto-search   | Required for auto-search & PDF download |
| Email address | Required for Unpaywall API | Your active email address               |
| OS            | Windows / macOS / Linux    | —                                       |

## Getting Started

### 2.1 How to Open the Tool

#### 1 Save the File

Save `AFM_Journal_Search_Tool_EN.html` to your computer.

#### 2 Open in Browser

**Drag and drop** the HTML file into a Chrome or Edge browser window, or right-click → Open with → Chrome.

#### 3 Ready to Use

All features are immediately available with no login or installation required.

#### ⚠ Important

Double-clicking the file may open it in your default browser (e.g., Internet Explorer), which may not fully support all features. Always open in **Chrome or Edge** for the best experience.

### 2.2 Interface Layout

The interface consists of a **top bar**, a **left sidebar menu**, and a **main content area**.

| Section            | Menu Items                                             | Purpose                         |
|--------------------|--------------------------------------------------------|---------------------------------|
| Workflow           | Keywords / Search Filters / Search Links               | Search preparation              |
| Journal Management | Auto Registration / Manual Registration / Journal List | Journal collection & management |
| Finish             | ZIP Export / Write Process                             | Output & reference              |

Click the `Dark Mode` button in the top-right to toggle dark/light theme.

## Workflow — Step-by-Step Guide

### STEP 1 | Keyword Setup

Click **Keywords** in the left sidebar. Keywords selected here are used for all subsequent searches and URL generation.

#### Selecting Preset Keywords

**30 preset AFM-related keywords** are provided. Active keywords are highlighted in blue. Click to toggle on/off.

##### ✦ Default Active Keywords

atomic force microscopy · tapping mode AFM · tip-sample interaction · cantilever dynamics · Q factor cantilever

#### Adding Custom Keywords

Type a keyword in the input field and press `Enter` or click **+ Add**. Refer to the Park Systems AFM-modes learning page for mode-specific keywords.

##### 🔗 Recommendation

Enter keywords in English. All academic DB searches operate on English-based queries. Example: `piezoresponse force`

`microscopy` , `Kelvin probe surface potential`

### STEP 2 | Search Filter Configuration

Click **Search Filters** in the sidebar.

##### 📅 Published After

Only search journals published after this year. Default: 2010

##### 📊 Min Impact Factor

Reference value for registration. IF is fetched automatically via Unpaywall in auto-search.

##### 📄 Min Citations

Reference threshold for filtering. Citation count is fetched automatically via CrossRef.

##### 🔢 Result Count

Number of results CrossRef API returns. Default: 20. Top 10 are PDF-verified.

#### Unpaywall API Email (Required)

Enter your email address to use the Auto Registration feature. The Unpaywall API is completely free and requires no API key. Your email is used only as a request identifier per their policy.

### STEP 3 | Academic Database Search

Click **Search Links** in the sidebar, or click **Generate Search Links** → from the Filters screen. Active keywords and year filters are automatically embedded into each DB's search URL.

| Database                 | Specialization           | Sci-Hub |
|--------------------------|--------------------------|---------|
| Google Scholar           | Broadest academic search | —       |
| ScienceDirect (Elsevier) | Elsevier journals        | ✓       |

|                      |                           |   |
|----------------------|---------------------------|---|
| Springer / Nature    | High-IF Nature family     | ✓ |
| ACS Publications     | American Chemical Society | ✓ |
| Wiley Online Library | Wiley journals            | ✓ |
| PubMed               | Bio/nano field            | — |

#### ⚠ **Paywalled Journals**

If you don't have institutional access, click the **Sci-Hub** button and enter the article's DOI to access the PDF.

## STEP 4 | Auto Journal Registration

Click **Auto Registration** in the sidebar. This feature automates the entire process from search to registration using CrossRef and Unpaywall APIs.

### 1 Verify Email

Confirm that your Unpaywall API email has been entered in the Search Filters tab.

### 2 Click Start Auto Search

The following runs automatically:

- ① CrossRef API → searches top results by relevance for active keywords
- ② Top 10 journals → each verified for PDF availability via Unpaywall API
- ③ Each journal card shows  PDF Available or  No PDF (Paywalled)

### 3 Choose Registration Method

- **Register All with PDF** — bulk register only open-access journals
- **Register All** — register all 10 regardless of PDF availability
- **+ Register** (per card) — selectively register individual journals

#### Auto-Filled Fields

Title · Author · Year · Journal name · Citations · DOI · PDF URL · Open Access status · Physics/Circuit content (keyword auto-detection) · Quality Score

## STEP 5 | Manual Registration via DOI

Click **Manual Registration** in the sidebar. Register journals found through manual browsing, or auto-fill via DOI.

### Auto-fill by DOI

#### 1 Enter DOI

Enter the paper's DOI from the academic DB.

Example: `10.1038/s41565-021-00844-6` or `https://doi.org/10.xxxx/...`

#### 2 Click Fetch

Metadata from CrossRef and the PDF URL from Unpaywall are automatically fetched and fill the form.

#### 3 Add Remaining Info & Register

Manually enter equation count (count from the Introduction section), notes, etc. The live quality score preview updates as you type. Click **+ Register Journal**.

## STEP 6 | Journal List Management

Click **Journal List** in the sidebar. All registered journals are displayed and manageable here.

#### Sort Options

Sort by score · citations · IF · year. Default: highest quality score first.

#### ☒ Selection Control

Checkbox on each card controls ZIP inclusion. Select All / Deselect All buttons provided.

#### PDF Quick Access

Journals with PDF URLs have a PDF button at the bottom of their card for instant access.

#### Statistics Summary

Total journals · selected · has PDF · average score displayed at the top.

## STEP 7 | ZIP Export

Click **ZIP Export** in the sidebar.

### 1 Enter AFM Mode Name

Example: `Tapping_Mode` , `PFM` , `KPFM` — used in the output filename.

### 2 Select Files to Include

Each of the 6 file types and PDF folder can be toggled individually. All are selected by default.

### 3 Click ↓ Download ZIP

If PDFs are included, each PDF is downloaded sequentially before compression. This may take several minutes depending on the number of journals.



## Journal Quality Score System

When a journal is registered, a quality score of **0-100** is automatically calculated using four criteria.

### Citations

40 pts

Max condition: 500+ citations — Score =  $\min(\text{citations}/500, 1) \times 40$

### Impact Factor

30 pts

Max condition:  $IF \geq 20$  — Score =  $\min(IF/20, 1) \times 30$

### Equations in Introduction

20 pts

Max condition: 20+ equations — Score =  $\min(\text{equations}/20, 1) \times 20$

### Physics included / Circuit & signal processing included

10 pts

Physics +5 pts, Circuit/signal processing +5 pts bonus. Auto-detected or manually selected.

#### ✦ Auto-Detection Keywords

**Physics:** van der Waals, tip-sample, cantilever, Hamaker, Hertz, DMT, JKR, oscillation, force

**Circuit:** lock-in, PID, feedback, amplifier, signal processing, phase detection, controller

## Score Grade Reference

| Score Range | Grade           | Recommended Use                            |
|-------------|-----------------|--------------------------------------------|
| 70-100      | ● Excellent     | Primary reference journal for Introduction |
| 40-69       | ● Good          | Use as supplementary reference             |
| 0-39        | ● Review Needed | Verify further before use                  |

#### 🔍 How to Count Equations

Equation count cannot be fetched automatically. Open the paper's Introduction section and manually count the equations, then enter the number in the field. This is the most important AFM-specific quality criterion.

## ZIP Output Files Detail

The ZIP file is downloaded as `AFM_Manual_[ModeName]_[Date].zip`.

| Filename                   |      | Format   | Contents                                                                                                                         |
|----------------------------|------|----------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>journals.json</b>       | JSON | JSON     | Full metadata for all selected journals. Includes IF · citations · score · PDF URL · DOI. Machine-readable for downstream tools. |
| <b>report.html</b>         | HTML | HTML     | Browser-viewable journal quality comparison report with score, PDF links, and DOI table.                                         |
| <b>keywords.txt</b>        | TXT  | TXT      | List of keywords used and the full search query string.                                                                          |
| <b>search_links.txt</b>    | TXT  | TXT      | Auto-generated search URLs for all 6 databases. Ready for re-use.                                                                |
| <b>intro_template.md</b>   | MD   | Markdown | Chapter 1 Introduction draft template. Includes key equations (van der Waals, Hertz, harmonic oscillator, etc.).                 |
| <b>review_checklist.md</b> | MD   | Markdown | AFM expert review checklist covering physical validity and system accuracy verification.                                         |
| <b>pdf/ folder</b>         | PDF  | PDF      | Auto-downloaded open-access PDFs. Replaced with a URL text file if download fails due to CORS or access restrictions.            |

### 🔗 Using intro\_template.md

Open with a Markdown viewer (VS Code, Typora, Obsidian) or convert to Word/Google Docs. LaTeX equations are written in `$$...$$` format.

## API Reference for Auto Registration

### 🔍 CrossRef API

**Purpose:** Keyword-based journal search, DOI metadata lookup

**Cost:** Completely free

**API key:** Not required

**Speed:** Results within seconds

**Rate limit:** 50 requests/sec (auto-compliant)

### 📄 Unpaywall API

**Purpose:** Verify open-access PDF URL per DOI

**Cost:** Completely free

**API key:** Not required (email only)

**Email purpose:** Abuse prevention identifier

**Rate limit:** 100,000 requests/hour

### ✉ Email Privacy

The email you enter is sent only to the Unpaywall API server and is not stored anywhere in the tool. Unpaywall does not use your email for marketing purposes.

## Limits of Open-Access PDF

Unpaywall only provides legally open-access PDFs. Subscription-required (paywalled) journal PDFs are not provided. In those cases:

- Access through your institutional library account, or
- Click the Sci-Hub button and enter the DOI, or
- Email the corresponding author to request a copy

## Frequently Asked Questions

### Q1. Can I use this tool without an internet connection?

Keyword setup, filter adjustment, manual journal registration (typed input), journal list management, and ZIP generation (excluding PDFs) all work offline. However, Auto Registration (CrossRef/Unpaywall API), DOI auto-fill, and PDF download require an internet connection.

### Q2. Auto search returns no results.

① Confirm your email is entered in the Search Filters tab. ② Verify that keywords are selected (highlighted in blue). ③ Check your internet connection. ④ Try broader or different keywords.

### Q3. Some journals show X No PDF.

This means the journal is paywalled — Unpaywall cannot provide its PDF. Use the Sci-Hub button with the DOI, or access through your institutional account.

### Q4. ZIP download is very slow.

When PDFs are included, each PDF is downloaded sequentially, so time scales with the number of journals. If PDFs are not needed, uncheck "pdf/ folder" in the ZIP Export screen for much faster generation.

### Q5. Do registered journals disappear when I close the browser?

Yes. The current version stores data only in browser session memory. Always export to ZIP before closing. All data is preserved in the `journals.json` file inside the ZIP.

### Q6. Can I set dark mode as the default?

Click the 🌙 Dark Mode button in the top-right to switch to dark theme. The setting is saved to browser local storage and persists on next launch.

## Next Version Roadmap

The following two major features are planned for the next version after v2.0.

### REVISION 1

#### Advanced Journal Grading Rules

- Differentiate equation weight by type (basic vs. core physics equations)
- Add field relevance index (auto-detect AFM topic alignment)
- Citation context analysis — distinguish core vs. incidental citations
- Tiered Impact Factor scoring with finer grade intervals
- Recency index — incorporate recent 5-year citation ratio

### REVISION 2

#### AI-Assisted Introduction Drafting from Registered Journals

- Auto-extract key sentences and summaries from registered journals
- Map content to the existing Introduction section structure
- Auto-map equations and references and generate a draft
- User review → feedback → regeneration workflow
- Export final draft as Markdown or Word document

#### ✦ Contact & Feedback

If you encounter issues or have suggestions while using the tool, please contact your Park Systems representative. User feedback is directly incorporated into the next version.